

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering



LAB REPORT

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Experiment No. : 1
Experiment Name : Identification of the Problems of a Targeted Information System

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Chapter 1

Identification of the Problems of a Targeted Information System

1. Experiment Information

Experiment Name: Identification of the Problems of a Targeted Information System

Selected Organization: RUET ICT Cell

Selected System: RUET POC app

2. Introduction

Information systems are designed to collect, process, store, and distribute information for decision-making and operational control in an organization. Before developing or improving such systems, it is essential to identify the existing problems, inefficiencies, or gaps within the targeted information system.

In this experiment, the selected organization is the RUET ICT Cell, and the target system is the RUET POC app. The system is analyzed to identify core elements, characteristics, and type, and the findings are mapped onto SDLC phases to understand how the system evolves from analysis to implementation and maintenance.

2.1 Organizational Structure of RUET ICT Cell

RUET ICT Cell is the central Information and Communication Technology department responsible for digital infrastructure, system administration, and IT service delivery across Rajshahi University of Engineering & Technology. The organizational hierarchy is as follows:

1. Director

- Dr. Md. Samiul Habib — Overall supervision of ICT operations and strategic planning.

2. Senior System Analysts

- S Mustafa Kamal — System analysis, infrastructure planning, and technical consultation.
- Engr. Shihan Arafin — System analysis, POC development oversight, and technical architecture.

3. Technical Development and Support Team

- Engr. Md. Julfikar Islam (Assistant Computer Programmer) — POC app backend development, Core PHP, MySQL.
- Md Ashif Al Amin (Assistant Network Engineer) — Network infrastructure, server maintenance, hosting administration.
- Md. Shahnewaz Uddin (Lab Assistant) — Technical support, data entry, system testing, and user assistance.

The POC development team (led by the Senior System Analysts and supported by the Assistant Computer Programmer and Network Engineer) is responsible for the RUET POC app development and maintenance. All staff report to the Director, Dr. Md. Samiul Habib.

3. Requirements

Knowledge Requirements

- Basic understanding of information system components (hardware, software, people, data, process).
- Familiarity with SDLC models (Waterfall, Iterative, Agile, etc.).
- Understanding of information system types (Transaction Processing System, MIS, DSS, etc.).

Materials / Tools Required

- Access to a real or simulated organizational information system.
- Internet and documentation tools (Google Docs, Word, Excel, or any CASE tool).
- Presentation and diagram tools (Lucidchart, Draw.io, or MS Visio).

4. Procedure

1. System Selection

- Selected organization: RUET ICT Cell.
- Selected system: RUET POC app.
- Collected background information on the app objective, intended users, and usage scope inside RUET.

2. Identify System Elements

- Hardware: Android smartphones (end users), desktops (admins), BdRen C panel hosting server.
- Software: RUET POC app (front-end mobile app), Core PHP backend, PHP MySQL, JS. (Version 2 will use Laravel, MySQL, jQuery, Tailwind CSS).
- People: Admin users (directed to POC), Teachers, Officers. Students are not focused on yet. 3rd/4th class employees are currently a challenge to include.
- Data: Name, designation, department, first joining date, employee ID, office location (building → floor → room), mobile number, IP phone, office mail, personal mail, and a 300x300 picture.
- Processes: Data input from poc.ruet.ac.bd, categorized into Teacher, Officer, and Staff modules. Output fed directly into the mobile app.

3. Classify the Information System

- System type: Central Database/Contact Directory and Transaction Processing System (TPS) initiating automation in RUET.
- Key characteristics: Single centralized SQL database handling all services. Currently relies on full online operation, with future attempts at offline functionality (potentially via Firebase syncing).

4. Analyze System Problems

- Identified issues existing before the POC: lack of personal mail usage, over-reliance on institutional mail, and identity/domain/hostname mapping issues.
- Identified current inclusion challenges (low-level staff integration) and technical debts (handling old uploaded photos without facial recognition, lack of NoSQL adaptability).

5. Apply SDLC Framework

- Mapped observations to Requirement Analysis (employee inclusion), Design (UI polishing, offline syncing), and Implementation (backend upgrade to Laravel/Tailwind).

6. Document and Report

- Prepared a structured report outlining current infrastructure, data fields collected, current DB limits, and targeted future expansions (Guest house booking, post-grad application).

5. Observed Findings

5.1 System Elements Identified

- Hardware: BdRen C panel server, user mobile devices.
- Software: Core PHP, MySQL, JavaScript, POC Android App (iOS app exists but unreleased).
- People: Teachers, Officers, Admins. (Not focused: Students. Challenging to include: 3rd/4th class physical workers).
- Data: Detailed contact fields (Name, Dept, Emp ID, Office Mail, Personal Mail, 300x300 Pic) and exact Office Location hierarchy.
- Processes: Data input via web (poc.rueta.ac.bd), categorized modularly, stored in one central SQL database, and fed to the app.

5.2 System Type and Characteristics

- System type: Baseline Automation and Central Directory System.
- Characteristics: Strict relational (SQL) structure, currently full online dependency, module-based categorization.

5.3 Key Problems Identified

- **Inclusion Gap:** Hard to cleanly integrate low-level (3rd/4th class) employees into the central digital system.
- **Technical Debt in Data:** Difficulties in managing and verifying older uploaded photos manually.

- **Architectural Limit:** Heavy dependency on continuous online connection; no built-in offline caching yet.
- **Historical Issues Fixed:** Eliminated ambiguity of institutional vs. personal mail and domain/emp-ID tracking.

5.4 SDLC-Based Mapping of Problems

- **Requirement Analysis:** Need for mechanisms to include low-level employees and integrate future modules like Guest House Booking and Post-Grad Application.
- **Design:** Future need for proper syncing between online/offline states (exploring Firebase). Need for 'Search' and 'Add to Favorite' UI options.
- **Implementation:** Upgrading the core backend stack. Version 2 will move to Laravel, MySQL, jQuery, and TailwindCSS for better modular future-proofing.
- **Testing:** Need algorithmic processing/testing for facial recognition to clean up the photo mismatch issue.
- **Maintenance:** Managing the single SQL database for all services as the system scales up.

6. Precautions

- **Data Confidentiality:** Employee personal and contact information must be handled under RUET ICT Cell confidentiality protocols. No raw database dumps or raw queries were disclosed in this analysis.
- **Authentic Domain Knowledge:** Claims about infrastructure, data handling, and future roadmap are based on direct dialogues with the RUET POC development team and system documentation.
- **Organizational Scope:** Analysis covers only the current RUET POC system (emp POC app and backend infrastructure). Future versions (v2) and planned integrations (Guest house, post-grad) are in scope for architectural discussion only.
- **System Boundary:** Analysis excludes student-focused modules (as they are not yet in scope) and focuses strictly on staff/faculty directory and contact consolidation.

7. Conclusion/Discussion

This experiment successfully analyzed the RUET POC system and identified critical architectural and operational challenges aligned with real-world system development practices.

Key Insights

The RUET POC system represents an important first-step automation effort at RUET by centralizing staff contact and directory information. The single-database SQL design ensures data consistency and eliminates historical ambiguities around employee identity and email assignment. However, the analysis revealed four major problem areas:

1. **Inclusion Equity Issue:** The system currently captures Teachers and Officers well but struggles with low-level (3rd/4th class) employees. This represents a fairness/accessibility design flaw in the Requirement Analysis phase.
2. **Offline-First Architecture Gap:** The current full-online dependency is fragile in poor-network environments. The Design phase must address offline caching and Firebase-based syncing.
3. **Data Quality and Verification:** Manual handling of legacy photos creates a bottleneck. Implementation of facial recognition in the Testing/Maintenance phases is a pragmatic long-term solution.
4. **Technology Stack Maturity:** The move to Laravel/Tailwind in v2 is sound, but requires careful migration planning and backward compatibility with the existing SQL schema.

Recommendations

1. **Short-term:** Develop an inclusion strategy for low-level staff (e.g., simplified registration flow, phone-based verify option).
2. **Medium-term:** Prototype offline caching using React Native or Flutter + Firebase to test robustness.
3. **Long-term:** Implement facial recognition API integration to automate photo verification and reduce manual QA overhead.

8. References

1. RUET ICT Cell. (2026). RUET POC System Architecture Documentation [Unpublished internal document].
2. BdRen (Bangladesh Research and Education Network). (2025). C-Panel Hosting Infrastructure Guidelines. *BdRen Technical Documentation*.
3. Firebase Documentation. (2025). Real-time Database Synchronization and Offline Support. Retrieved from <https://firebase.google.com/docs/database/>
4. Laravel Official Documentation. (2025). Backend Framework and Database Abstraction Layers. Retrieved from <https://laravel.com/docs/>
5. Sommerville, I. (2015). *Software Engineering* (10th ed.). Pearson Education. [SDLC reference].

9. Appendix

A. Context Diagram (Placeholder)

[A visual representation of RUET POC data flow showing:]

- External actors (Teachers, Officers, Staff, Admin).
- Primary system boundary (POC app and central SQL database).
- Data inputs (poc.ruet.ac.bd portal) and outputs (mobile app notifications, admin dashboards).

B. Data Flow Diagram (Placeholder)

[A detailed DFD showing:]

- Three module processing streams (Teacher, Officer, Staff modules).
- Data transformation steps (input validation, categorization, storage).
- Database interactions (read/write operations, synchronization).

C. System Technical Details

Table 1.1: RUET POC Technical Stack Summary

Component	Current	Future (v2)
Frontend Mobile	Core PHP + JS, Android app	(Planned upgrade)
Backend/API	Core PHP + MySQL	Laravel + MySQL
UI Framework	Custom	TailwindCSS
Hosting	BdRen C-Panel	(Continued or upgraded)
Offline Sync	None (fullonline)	Firebase (planned)
Authentication	Role-based (Admin/Faculty)	Planned expansion

D. Interview/Consultation Notes

- Discussed system evolution roadmap with RUET ICT Cell: Priority order for v2 features is (1) offline support, (2) UI redesign, (3) new module integration.
- Noted that 3rd/4th class employee inclusion is a policy-level decision as much as a technical one.
- Firebase integration for offline-online syncing is already under evaluation by the development team.